

ORIGINAL ARTICLE

Clinical Characteristics and Outcomes of Hypertrophic Cardiomyopathy Associated With Thin Filament Variants: Insights From the SHaRe Registry

Alessia Argirò¹, MD, PhD; Joshua Meisner², MD, PhD; Eric Adler³, MD, PhD; Edoardo Bertero⁴, MD, PhD; Niccolò Maurizi⁵, MD; Brian Claggett⁶, PhD; Neal K. Lakdawala⁷, MD; Anjali Owens⁸, MD; Thomas D. Ryan⁹, MD, PhD; Erin M. Miller¹⁰, MS, CGC; Sara Saberi¹¹, MD, MS; Henning Bundgaard¹², MD, DMSc; Anna Axelsson Raja¹³, MD, PhD; Joseph W. Rossano¹⁴, MD, MS; Kimberly Lin¹⁵, MD; Francesca Girolami¹⁶, CGC; Michelle Michels¹⁷, MD, PhD; Peter Paul Zwetsloot¹⁸, MD, PhD; Dominic Abrams¹⁹, MD; Rachel Lampert²⁰, MD; John C. Stendahl²¹, MD, PhD; James Ware²², PhD; Victoria N. Parikh²³, MD; Euan Ashley²⁴, MD, PhD; Jodie Ingles²⁵, PhD; Belinda Gray²⁶, PhD; Romi L. Castillo²⁷, PhD; Jil Tardiff²⁸, MD, PhD; Lia Crotti²⁹, MD, PhD; Alexandre Pereira³⁰, MD, PhD; Carolyn Y. Ho³¹, MD; Sharlene Day³², MD, PhD; Adam Helms³³, MD, PhD*; Iacopo Olivetto³⁴, MD*



BACKGROUND: Conflicting evidence exists about whether sarcomeric hypertrophic cardiomyopathy (HCM) caused by genes encoding thin filament proteins (*TNNT2*, *TNNI3*, *ACTC*, *TPM1*, *TNNC1*; thin filament HCM) has a distinct clinical presentation and natural history from disease caused by the more commonly involved thick filament genes (*MYH7*, *MYBPC3*, *MYL2*, *MYL3*; thick filament HCM).

METHODS: Retrospective analysis of the Sarcomeric Human Cardiomyopathy Registry.

RESULTS: Four hundred eighty-eight patients with thin filament HCM were compared with 3494 patients with thick filament HCM. After controlling for age at diagnosis, sex, and proband status, thin filament HCM was associated with a higher risk of composite ventricular arrhythmias (hazard ratio, 1.48 [95% CI, 1.09–2.01]), cardiac arrest (hazard ratio, 2.02 [95% CI, 1.33–3.06]), and heart transplant or left ventricular assist device implantation (hazard ratio, 2.18 [95% CI, 1.3–3.65]). Younger age at diagnosis was associated with an increased risk for the ventricular arrhythmia composite outcome in thin and thick filament HCM (hazard ratio, 0.93 [95% CI, 0.91–0.94]; hazard ratio, 0.93 [95% CI, 0.93–0.94]; age reported as 5-year increment). Within specific thin filament genes, an increased risk of arrhythmic events or heart transplant/left ventricular assist device implantation was evident for variants in *TNNT2* amino acids 92 to 94 (odds ratio versus other *TNNT2* variants; odds ratio, 3.02 [95% CI, 1.17–7.49]) and *TNNI3* amino acids 191 to 210 (odds ratio versus other *TNNI3* variants; odds ratio, 3.5 [95% CI, 1.58–8.02]).

CONCLUSIONS: Thin filament HCM seems to carry a higher risk for ventricular arrhythmias and advanced heart failure compared with thick filament HCM. Integrating gene-level and intragene-level information may improve risk stratification in patients with HCM.

Key Words: atrial fibrillation ■ cardiomyopathies ■ genes ■ genetics ■ patients

Hypertrophic cardiomyopathy (HCM) is the most common inherited cardiac condition and is defined by otherwise unexplained left ventricular (LV)

hypertrophy.¹ A pathogenic/likely pathogenic (P/LP) sarcomeric variant may be identified in up to 60% to 70% of patients and is associated with an increased

Correspondence to: Alessia Argirò, MD, PhD, Cardiomyopathy Unit, University of Florence, Largo Brambilla 3, 50141 Florence, Italy. Email alessia.argiro@unifi.it
This manuscript was sent to Julie De Backer, Guest Editor, for review by expert referees, editorial decision, and final disposition.

*A. Helms and I. Olivetto contributed equally.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCGEN.126.005696>.

© 2026 American Heart Association, Inc.

Circulation: Genomic and Precision Medicine is available at www.ahajournals.org/journal/circgen

Nonstandard Abbreviations and Acronyms

HCM	hypertrophic cardiomyopathy
HR	hazard ratio
IQR	interquartile range
LV	left ventricle/ventricular
LVAD	left ventricular assist device
P/LP	pathogenic/likely pathogenic
SCD	sudden cardiac death
SHaRe	Sarcomere Human Cardiomyopathy Registry
VUS	variant of uncertain significance

risk of adverse events compared with nonsarcomeric HCM.² The most frequent sarcomere variants are found in thick filament genes, which encode for myosin binding protein C (*MYBPC3*, ~55%) and beta myosin heavy chain (*MYH7*, ~30%).² However, up to 15% of cases of sarcomere-related HCM are caused by variants in genes encoding thin filament proteins, including tropomyosin (*TPM1*), actin (*ACTC1*), and the troponin complex (*TNNT2*, *TNNI3*, and *TNNC1*).

Variants in thin filament genes result in conformational changes that lead to altered stoichiometry (relative protein proportions and spatial organization) and dynamic cooperativity (protein-protein interactions), a mechanism that may be distinct from thick filament variants.³ Indeed, prior analyses have identified distinct clinical profiles characterized by more severe diastolic dysfunction and a greater propensity for advanced heart failure.^{4,5}

However, due to the rarity of thin filament HCM, it has been difficult to consistently determine genotype-phenotype correlations that distinguish thick and thin filament HCM, with small cohort studies showing conflicting results.^{5,6} In an effort to resolve this discrepancy, we leverage the multicenter international SHaRe (Sarcomere Human Cardiomyopathy Registry) to compare the clinical characteristics and outcomes of patients with thin and thick filament HCM.

METHODS

Study Design

SHaRe is an international, multicenter observational platform with both retrospective and prospective components, collecting data from 13 expert, high-volume HCM referral centers. Consecutive patients who agree to participate are enrolled during routine visits at dedicated HCM clinics. Enrollment into the registry began in 2008, while genetic testing has been performed as part of routine clinical care at participating centers since 1992. Participating sites are specialized cardiomyopathy referral centers; therefore, referral bias toward more severe phenotypes cannot be excluded. The registry includes

both probands and relatives identified through family cascade screening, and centers maintain longitudinal databases capturing clinical, genetic, and outcome data on patients with HCM and their families.²

Institutional review board and ethics committee approvals were obtained at each SHaRe site, and the subjects gave informed consent. The data that support the findings of this study cannot be shared due to privacy regulations. The methods used in the analysis are available from the corresponding author upon reasonable request.

The diagnosis of HCM was made by experienced clinicians and based on the presence of a maximal LV wall thickness of ≥ 15 mm (or body surface area-adjusted Z score >2 for children) or ≥ 13 mm in the presence of first-degree relatives with unequivocal disease or positive genetic testing, taking into consideration medical and family history and genotype.^{1,7}

Cardiac dimensions and function have been evaluated through echocardiographic measurements reported on clinical visits performed at experienced HCM centers. Genetic testing was performed at each site using different platforms based on availability and clinical practice.

Genetic testing was focused on the 9 sarcomere variants associated with HCM (*MYBPC3*, *MYH7*, *TNNT2*, *TNNI3*, *TNNC1*, *TPM1*, *MYL2*, *MYL3*, and *ACTC1*). Variants were classified as P, LP, variant of uncertain significance (VUS), or likely benign/benign at each site using American College of Medical Genetics and Genomics and Association for Molecular Pathology criteria. Variants with ambiguous or discordant classifications underwent additional systematic review by a subgroup of SHaRe investigators for resolution.² In our population, sarcomere VUS variants have a known association with HCM severity consistent with biological relevance.² This study utilized P/LP variants; additional analyses of PLP+VUS and VUS were performed individually to compare associations identified within the primary cohort of P/LP variants. Thus, patients with P/LP in *MYBPC3*, *MYH7*, *MYL2*, or *MYL3* were classified as thick filament HCM, and patients with P/LP in *TNNT2*, *TNNI3*, *TNNC1*, *TPM1*, or *ACTC1* were classified as thin filament HCM. The *TNNT2* C.832C>T low penetrance sarcomere variant has not been included in this analysis because it was classified in SHaRe as likely benign.⁸

Patients with potentially pathogenic variants in genes encoding nonsarcomere cardiac proteins or others related to HCM phenocopies such as Danon disease, Fabry disease, or cardiac amyloidosis were excluded. Patients with complex genotypes, including those with both thin and thick filament variants, were excluded.

End Points

The following outcomes were evaluated²:

- Ventricular arrhythmia composite outcome: first occurrence of sudden cardiac death, resuscitated cardiac arrest, or appropriate implantable cardioverter defibrillator therapy.
- The overall composite outcome included the following: first occurrence of any component of the ventricular arrhythmia composite outcome, first occurrence of heart transplant, left ventricular assist device (LVAD) implantation, New York Heart Association III/IV symptoms, atrial fibrillation, stroke, or all-cause mortality.

- Cardiac arrest
- Heart transplant/LVAD implantation.
- Atrial fibrillation.
- All-cause death.

Statistical Analysis

We analyzed SHaRe data through February 2025. Continuous variables are presented as median (interquartile range [IQR]), and categorical variables are presented as counts and percentages. Between-group comparisons were evaluated using the Wilcoxon rank-sum test and χ^2 tests as appropriate.

Time-to-event analyses were performed using age as the time scale, with time origin set at birth and events defined by age at occurrence. Patients who did not have the outcome of interest were censored at the time of their last recorded follow-up in SHaRe. Patients missing data on the occurrence or timing of events were not included in the analyses of those outcomes. However, to maximize sample sizes across analyses, such patients were included in analyses of other outcomes for which data were available.

The Cox proportional hazards model was used to evaluate the association of baseline variables with the events of interest. Cox regression results are reported using adjusted hazard ratios (HRs) from multivariable analyses. Adjusted survival curves were estimated using a Cox proportional hazards model with covariates for proband status and sex, using the R package adjustedCurves. The proportional hazards assumption was tested based on Schoenfeld residuals. On an exploratory basis, to evaluate the presence of high-risk variants, a χ^2 test was performed to compare the rate of events across different variants, considering previously reported data.^{9–12} A logistic regression model was used to quantify the excess risk of events associated with specific high-risk variants based on the previous test. Statistical significance was defined as an $\alpha < 0.05$. All analyses were performed using R (version 4.3.1).

RESULTS

Genetic Background of the Study Population

Of 10940 patients with a primary diagnosis of HCM, 6793 underwent clinical genetic testing, in whom 325 had a P/LP or VUS in thin filament genes, compared with 3494 in thick filament genes (155 unique variants in thin filament genes and 787 in thick filament genes; Table S1). VUSs were more prevalent in thin-filament HCM than in thick-filament HCM ($n=163$, 33% versus $n=680$, 19%; $P < 0.01$). Among patients with thin filament HCM, P/LP variants were identified in 325 (*TNNI3* $n=139$, 43%; *TNNT2* $n=136$, 42%; *TPM1* $n=35$, 11%; *ACTC1* $n=9$, 3%; *TNNC1* $n=6$, 2%), and VUSs were identified in 163 patients (*TNNI3* $n=25$, 15%; *TNNT2* $n=55$, 34%; *TPM1* $n=46$, 28%; *ACTC1* $n=26$, 16%; *TNNC1* $n=11$, 7%). Variants were predominantly missense (*TNNI3* $n=153$, 93%; *TNNT2* $n=148$, 78%; *ACTC1* $n=31$, 89%; *TPM1* $n=72$, 89%; *TNNC1* $n=17$, 100%). The most commonly affected protein domains included the mobile domain for *TNNI3* ($n=80$, 49%) and the tropomyosin-binding site 1 for *TNNT2*

($n=103$, 54%; Tables S2 and S3; Figures S1 through S5). Among patients with thick filament HCM (P/LP variants: *MYBPC3* 66.5%, *MYH7* 31.8%, *MYL2* 1.4%, *MYL3* 0.3%; VUSs: *MYBPC3* 48.5%, *MYH7* 38.6%, *MYL2* 7.9%, *MYL3* 5%), the prevalence of missense variants was 29% for *MYBPC3*, 97% for *MYH7*, 89% for *MYL2*, and 90% for *MYL3*.

Clinical Characteristics of Patients With Thin-Filament Versus Thick-Filament HCM

In patients with thin-filament HCM, the median age at diagnosis was 36 (IQR, 19–53) years, and 167 (51%) of the patients were men, comparable to patients with thick-filament HCM (median age, 39 years; IQR, 24–51; men: $n=1625$, 58%, Table 1). Patients with thin-filament HCM were less likely to have a family history of HCM ($n=175$, 54% versus $n=1792$, 64%; $P < 0.01$), showed milder LV hypertrophy (maximal LV wall thickness, 16 mm; IQR, 13–21 versus 18 mm; IQR, 14–22; $P < 0.01$) and were less likely to have an obstructive phenotype ($n=74$, 27% versus $n=1065$, 42%, $P < 0.001$). As a marker of diastolic dysfunction, left atrial volume was greater in thin-filament HCM compared with thick-filament HCM (43 mL/m²; IQR, 33–59 versus 39 mL/m²; IQR, 29–51, $P=0.003$). Separate descriptive characteristics of patients with P/LP+VUS variants and VUS are reported in Tables S4 and S5.

Time-to-Event Analysis in Patients With Thin- and Thick-Filament HCM

In thin-filament HCM, the median time of follow-up from diagnosis was 7.6 (IQR, 2.9–13.9) years, representing a total of 3147 patient-years, while for thick-filament HCM, median follow-up time was 8.3 (IQR, 2.2–16.8) years for a total of 28813 patient-years.

Compared with patients with thick-filament HCM, patients with thin-filament HCM displayed a higher incidence of the ventricular arrhythmia composite outcome (3.09 [95% CI, 2.22–3.97] per 1000 person-years versus 2.14 [95% CI, 1.90–2.38] per 1000 person-years; $P=0.018$; Log-rank $P=0.012$, Figure 1; Table S6) and a higher incidence of cardiac arrest (1.73 [95% CI, 1.08–2.38] per 1000 person-years 0.87 [95% CI, 0.72–1.02] per 1000 person-years, $P < 0.001$; Log-rank $P < 0.001$). Patients with thin-filament genes also had a higher incidence of HT/LVAD implantation compared with patients with variants in thick-filament genes (incidence rate, 1.13 [95% CI, 0.61–1.66] per 1000 person-years versus 0.51 [95% CI, 0.39–0.63], $P=0.002$, Log-rank $P=0.001$). Sensitivity analyses were performed, including patients with P/LP+VUS variants, and showed similar findings (Table S7; Figure S6). Likewise, patients with thin-filament VUSs had a higher incidence of cardiac arrest compared with patients with VUSs in thick filament genes (Table S8; Figure S7). Survival curves adjusted for

Table 1. Baseline Characteristics of Patients With Pathogenic/Likely Pathogenic Variants in Thin and Thick Filament Genes

Variable	Thick filament, n=2814	Thin filament, n=325	P value
Age at primary diagnosis, median (IQR), y	38.8 (24.0–51.4)	35.5 (19.4–53.4)	0.2*
Race, n (%)			0.11†
Asian	90/2579 (3.5%)	17/295 (5.8%)	
Black	86/2579 (3.3%)	7/295 (2.4%)	
White	2403/2579 (93%)	271/295 (92%)	
LVEF at first evaluation, median (IQR), %	65.0 (60.0–70.0)	64.0 (60.0–70.0)	0.7*
Maximal LV wall thickness, median (IQR), mm	18.0 (14.0–22.0)	16.0 (13.0–21.0)	<0.001*
Left atrial volume, median (IQR), mL/m ²	38.6 (29.1–51.1)	43.3 (33.0–58.7)	0.003*
LVOT gradient, median (IQR), mm Hg	10.1 (8.5–30.0)	10.1 (7.0–10.1)	<0.001*
ESCRI risk score	2.4 (1.5–3.8)	2.2 (1.4–3.5)	0.2*
Pediatric, n (%)	87/2814 (3.1%)	11/325 (3.4%)	0.8†
Male sex, n (%)	1625/2814 (58%)	167/325 (51%)	0.028†
Proband, n (%)	2200/2814 (78%)	277/325 (85%)	0.003†
Hypertension, n (%)	818/2814 (29%)	61/325 (19%)	<0.001†
Family history of SCD/ESC, n (%)	350/2814 (12%)	41/325 (13%)	>0.9†
Family history of HCM, n (%)	1792/2814 (64%)	175/325 (54%)	<0.001†
NYHA class at first visit, n (%)			0.6†
1	1388/2418 (57%)	149/276 (54%)	
2	744/2418 (31%)	95/276 (34%)	
3	270/2418 (11%)	31/276 (11%)	
4	16/2418 (0.7%)	1/276 (0.4%)	
Follow-up time, median (IQR), y	7.6 (2.9–13.9)	8.3 (2.2–16.8)	0.3*
Age at atrial fibrillation, median (IQR), y	50.0 (36.8–61.4)	48.0 (33.1–60.6)	0.082*
Age at cardiac arrest, median (IQR), y	52.7 (38.2–63.9)	50.0 (33.1–64.5)	0.044*
Age at ICD implantation, median (IQR), y	49.2 (35.1–61.1)	47.7 (27.7–61.6)	0.13*
Age at ICD complication, median (IQR), y	52.8 (38.2–64.0)	50.6 (33.4–64.5)	0.090*
Age at ventricular arrhythmia composite outcome, median (IQR), y	52.4 (37.8–63.7)	50.0 (32.9–64.3)	0.040*
Age at composite overall outcome, median (IQR), y	49.1 (36.0–60.3)	46.8 (30.7–59.8)	0.033*
Age at HT/LVAD implantation, median (IQR), y	53.0 (38.4–64.1)	50.6 (33.6–64.5)	0.078*
Age at sudden cardiac death, median (IQR), y	53.0 (38.5–64.2)	50.6 (33.6–64.7)	0.080*

ESC indicates European Society of Cardiology; HCM, hypertrophic cardiomyopathy; ICD, implantable cardioverter defibrillator; IQR, interquartile range; LVAD, Left ventricular assist device; LVEF, left ventricular ejection fraction; LVOT, left ventricular outflow tract; NYHA, New York Heart Association; and SCD, sudden cardiac death.

*P values were calculated using Wilcoxon signed-rank test.

†P values were calculated using χ^2 test.

sex and proband status are reported in the [Supplemental Material \(Figures S8 through S10\)](#).

Exploratory Analysis Compared the ESC Risk-Sudden Cardiac Death Score Between Patients Who Did and Did Not Develop the Ventricular Arrhythmia Composite Outcome

Although patients with thick-filament HCM who developed the composite ventricular arrhythmic outcome had higher HCM risk-sudden cardiac death (SCD) scores than those who did not, HCM risk-SCD scores were similar between patients with thin-filament HCM regardless of arrhythmic outcome ([Table S9](#)). Furthermore, among

patients who developed the ventricular arrhythmia composite outcome, patients with thin filament had lower HCM risk-SCD scores compared with those with thick filament HCM.

Single-Variable and Multivariable Outcomes Analysis in Patients With Thin- and Thick-Filament HCM

After adjusting for age at diagnosis, sex, and proband status, patients with thin-filament HCM displayed a higher risk of the ventricular arrhythmia composite outcome (HR, 1.48 [95% CI, 1.09–2.01]), cardiac arrest (HR, 2.02 [95% CI, 1.33–3.06]), and HT/VAD (HR, 2.18

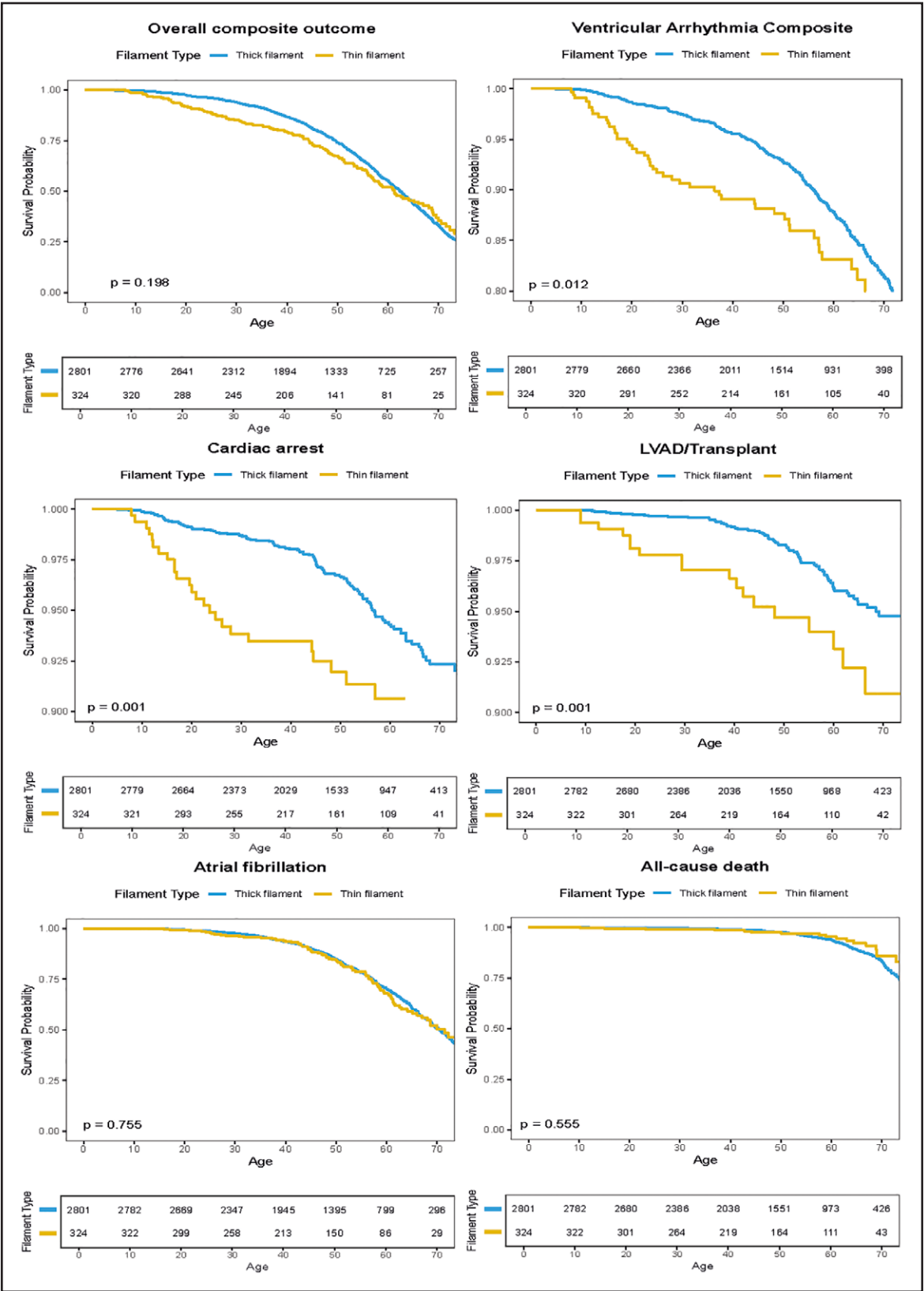


Figure 1. Kaplan-Meier curves of patients with pathogenic/likely pathogenic variants.
P values were calculated using the Log-rank test. LVAD indicates left ventricular assist device.

[95% CI, 1.30–3.65]; Table S10; Figure 2). The result was similar in the patients with P/LP or VUS variants (ventricular arrhythmia composite outcome HR, 1.37 [95% CI, 1.05–1.79], cardiac arrest HR, 2.09 [95% CI, 1.47–2.97], HT/VAD implantation HR, 1.7 [95% CI, 1.04–2.76]), although patients with VUS in thin-filament genes showed a higher risk of cardiac arrest compared with those with a VUS in thick filament genes (HR, 3.11 [95% CI, 1.54–6.26]; Figures S11 and S12).

In multivariable analysis, younger age at diagnosis was associated with an increased risk of ventricular arrhythmia composite outcome, atrial fibrillation, cardiac arrest, HT/LVAD implantation, and composite overall outcome in both thin-filament and thick-filament HCM (Table 2). Sex differences in outcome predictors were not evident in thin-filament HCM, although male sex was associated with a higher risk of the ventricular arrhythmia composite outcome and atrial fibrillation in thick-filament HCM, but a lower risk of HT/VAD implantation. Multivariable analysis for patients with P/LP+VUS is reported in Table S11.

There was no overall difference in outcome when individual thin filament genes were compared (Figures S13 and S14; Tables S12 and S13). Based upon occurrence rates of arrhythmia events in the most prevalent thin-filament genes and previous published data,⁹ an excess rate of adverse events was evident for variants affecting

amino acids 92 to 94 in *TNNT2* (odds ratio for the ventricular arrhythmia and VAD implantation/HT composite outcome versus the other *TNNT2* variants odds ratio, 3.02 [95% CI, 1.17–7.49]) and amino acids 191 to 210 in *TNNI3* (OR for the ventricular arrhythmia and VAD implantation/HT composite outcome versus the other *TNNI3* variants odds ratio, 3.5 [95% CI, 1.58–8.02]; Figures S15 and S16).

DISCUSSION

In this study of over 3000 patients with sarcomeric HCM, thin-filament HCM was associated with higher risk of developing life-threatening ventricular arrhythmias and advanced heart failure requiring HT or LVAD compared with thick-filament HCM. Notably, risk was the greatest in patients with a younger age at diagnosis, although no sex-related differences emerged. In agreement with previous studies, patients with variants in thin filament genes had less LV hypertrophy, larger left atrial volume, and were less likely to have obstructive physiology compared with thick filament HCM. While left atrial enlargement may reflect chronically elevated filling pressures, a comprehensive assessment of diastolic function was not performed in this study, and therefore no conclusions about restrictive physiology can be drawn from our data.

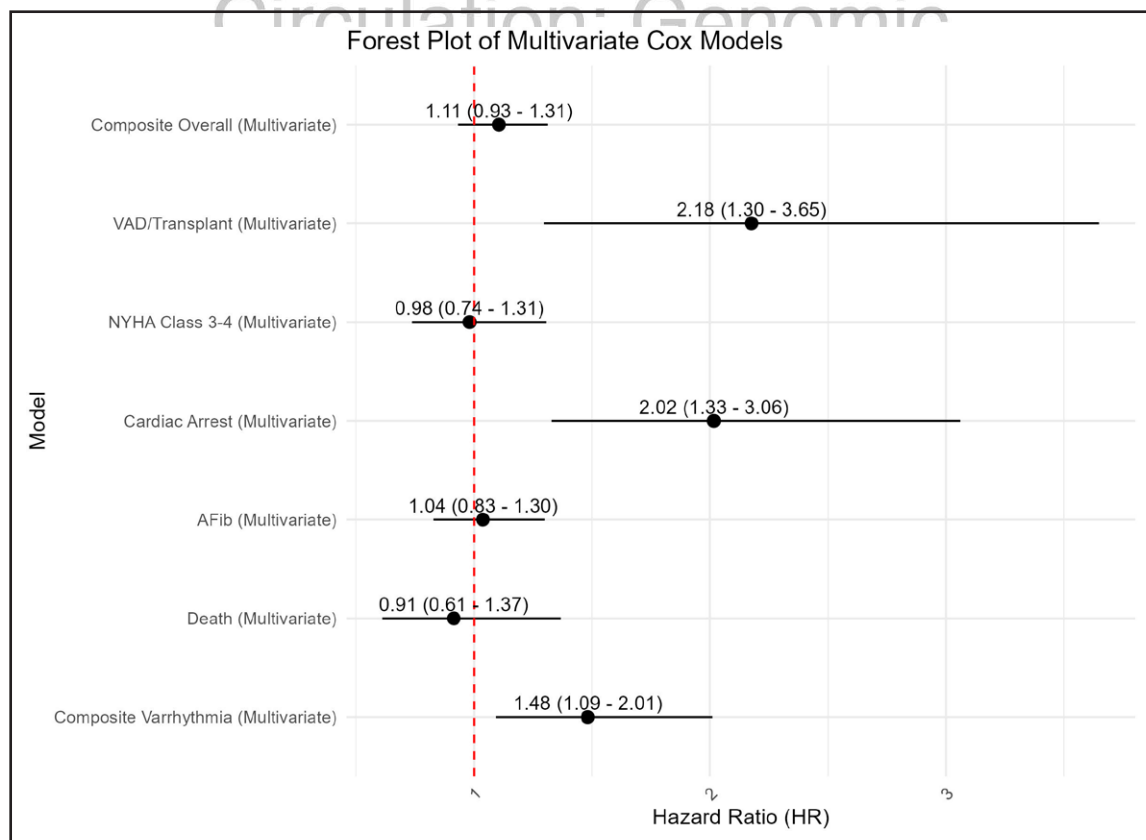


Figure 2. Cox multivariable model of patients with pathogenic/likely pathogenic in thin vs thick filament (adjusted for age at diagnosis, sex, and proband status).

Dashed line represents hazard ratio=1. NYHA indicates New York Heart Association; and VAD, ventricular assist device.

Table 2. Multivariable Analysis in Patients With P/LP in Thick and Thin Filament Genes

Outcome	Thin filament					Thick filament			
	Covariate	HR	Lower 95	Upper 95	P value	HR	Lower 95	Upper 95	P value
Composite ventricular arrhythmias	Age at primary diagnosis	0.92*	0.9*	0.94*	<0.01*	0.93*	0.93*	0.94*	<0.01*
	Male sex	1.01	0.57	1.79	0.978	1.4*	1.09*	1.78*	0.007*
	Proband status	1.01	0.44	2.32	0.973	1.34	0.97	1.86	0.079
Atrial fibrillation	Age at primary diagnosis	0.96*	0.94*	0.97*	<0.01*	0.95*	0.95*	0.96*	<0.01*
	Male sex	1.46	0.94	2.28	0.095	1.14	0.99	1.32	0.078
	Proband status	0.66	0.37	1.19	0.168	1.33*	1.1*	1.62*	0.004*
Death	Age at primary diagnosis	0.93*	0.9*	0.96*	<0.01*	0.95*	0.94*	0.96*	<0.01*
	Male sex	1.76	0.76	4.12	0.19	1.15	0.91	1.46	0.242
	Proband status	1.27	0.38	4.2	0.696	1.09	0.8	1.48	0.582
Cardiac arrest	Age at primary diagnosis	0.93*	0.9*	0.96*	<0.01*	0.94	0.93	0.95	<0.01
	Male sex	1.07	0.5	2.29	0.865	1.34	0.92	1.95	0.131
	Proband status	0.94	0.32	2.72	0.906	1.24	0.76	2.02	0.399
Heart transplant/LVAD implantation	Age at primary diagnosis	0.89*	0.85*	0.93*	<0.01*	0.9*	0.88*	0.91*	<0.01*
	Male sex	1.37	0.53	3.58	0.518	0.55*	0.34*	0.88*	0.013*
	Proband status	2.57	0.34	19.53	0.362	1.37	0.68	2.78	0.378
Composite overall	Age at primary diagnosis	0.94*	0.93*	0.95*	<0.01*	0.95*	0.94*	0.95*	<0.01*
	Male sex	0.97	0.69	1.35	0.848	1	0.89	1.12	0.968
	Proband status	0.85	0.54	1.34	0.479	1.3*	1.12*	1.51*	0.001*

Age is reported as 5-y increment. HR indicates hazard ratio; LVAD, left ventricular assist device; and P/LP, pathogenic/likely pathogenic.

Previous reports have described an association between thin filament variants and restrictive pathophysiology, which may overlap with primary restrictive HCM, particularly in children and adolescents.^{4,5} An analysis of the ClinVar database supports this genotype-phenotypic link: among variants specifically submitted with a restrictive cardiomyopathy phenotype, 99.5% were localized in thin filament genes (Tables S14 and S15).

While intergenic differences were limited in our thin filament population, an excess rate of arrhythmic events or heart transplant/LVAD implantation was evident for variants in amino acids 92 to 94 in *TNNT2* and amino acids 191 to 210 in *TNNI3*. This observation is in line with early family studies describing high arrhythmic risk associated with missense variants localized in the interaction sites of thin-filament proteins with other sarcomere components, such as R21C, R98Q, and R145T variants in *TNNI3* gene, and the R92W variant in *TNNT2*.^{9,13–15} Conversely, variants in other regions of *TNNT2* have been associated with favorable outcomes,⁹ highlighting an intrinsic diversity. Overall, our data provide robust support for earlier reports highlighting a distinct clinical profile associated with thin filament disease, compared with the more prevalent, classic model of thick filament HCM. This argues the importance of developing thin-filament-specific risk models.

The distinct profile of thin filament HCM reflects complex and diverse pathophysiological mechanisms, in view of the central role of the troponin complex in regulating

muscle contraction.¹⁶ Prior preclinical studies suggest that thin filament variants are associated with increased calcium sensitivity, which may impair relaxation and promote arrhythmogenicity.^{17–20}

Variants affecting the C-terminal region of cTnI and TNNT2 (92–94) in the TM-overlap domain may disrupt normal thin filament deactivation by impairing its regulatory interactions with actin and tropomyosin, leading to greater calcium sensitivity, delayed relaxation, persistent crossbridge formation, increased myocardial stiffness, and arrhythmic propensity.^{21–23}

Although it remains to be seen how generally applicable these proposed mechanisms will prove to be, robust coupled clinical, computational, and structural approaches represent both a strong framework for future study as well as identifying novel thin-filament targets for early intervention to alter the natural history of these complex disorders.

Our findings have potential clinical implications. The identification of thin filament gene variants associated with HCM should raise concerns about adverse outcomes and increased arrhythmic propensity, particularly in younger patients and in those with variants in sensitive regions of TNNT2 and TNNI3. With regard to disease progression, worsening of diastolic function and emergence of restrictive patterns should be sought, irrespective of systolic function, which tends to be preserved even at severe stages and may falsely reassure. With regard to arrhythmic risk, in the derivation and validation cohorts of

the HCM risk-SCD model,^{24,25} systematic genetic testing was not reported, and therefore the representation of patients with thin filament variants cannot be determined. However, thin filament HCM is typically characterized by lower LV wall thickness and a lower prevalence of LV outflow tract obstruction, 2 key variables included in the risk model. As a result, there is a plausible concern that the HCM risk-SCD score may underestimate arrhythmic risk in this subgroup, as supported by our data showing that, among patients who developed the ventricular arrhythmia composite outcome, those with thin filament HCM had lower HCM risk-SCD scores than those with thick-filament HCM.

Furthermore, in our cohort, HCM risk-SCD scores were similar between patients with thin filament HCM regardless of arrhythmic outcome, suggesting that conventional risk stratification parameters may not adequately discriminate risk in this subgroup. Accordingly, clinicians should be aware that arrhythmic risk may not be fully captured by the current model. Further studies are needed to evaluate the performance and calibration of existing risk scores in genetically defined subgroups and to determine whether genotype-specific approaches may improve risk stratification.

Male sex in our cohort was associated with a higher risk of ventricular arrhythmias and atrial fibrillation, and a lower risk of HT/VAD implantation in the thick filament subgroup. This aligns with prior HCM literature, where men tend to have greater arrhythmic risk, while women more often progress to heart failure.²⁶ However, the apparent difference between thick and thin filament subgroups should be interpreted cautiously. The larger size and higher event rate in the thick filament group provided greater statistical power, whereas the smaller thin filament group limited the detection of similar associations. Therefore, these findings likely reflect differences in sample size rather than true biological variation in sex-specific risk between subgroups.

Limitations of the present study include those inherent to retrospective, registry-based observational studies, including survivor bias and the fact that inferences about causality cannot be made. Furthermore, all participating centers are referral centers, which may introduce selection bias. The number of patients with thin filament HCM was relatively small, therefore the predictive power of Cox proportional hazard model may be reduced. Furthermore, we were unable to characterize diastolic function or quantify late gadolinium enhancement on cardiac magnetic resonance imaging, a parameter associated with heart failure and arrhythmic risk in HCM.^{1,7}

CONCLUSIONS

Thin filament HCM appeared to carry a higher risk for malignant arrhythmias and advanced heart failure compared with thick filament HCM. Arrhythmic/HF risk

was greatest for amino acids 92-94 in *TNNT2* and 191-210 in *TNNI3*. This study provides greater nuance in understanding the impact of genetic cause in HCM, and it supports the concept that integrating intra- as well as inter-gene analyses may improve risk stratification in patients with HCM. Further studies are needed to improve genotype-specific risk stratification.

ARTICLE INFORMATION

Received February 12, 2026; accepted July 9, 2026.

Affiliations

Cardiomyopathy Unit, University of Florence, Italy (A.A., N.M., I.O.). Department of Pediatrics (J.M.), Division of Cardiovascular Medicine, Department of Internal Medicine (S.S.), and Cardiovascular Medicine (F.G., A.H.), University of Michigan, Ann Arbor. UC San Diego Health, CA (E. Adler). Cardiovascular Disease Unit, Cardiac, Thoracic and Vascular Department, IRCCS Ospedale Policlinico San Martino, Genova, Italy (E.B.). Service of Cardiology, University Hospital of Lausanne, Switzerland (N.M.). Cardiovascular Division, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA (B.C., N.K.L., C.Y.H.). Perelman School of Medicine (A.O.) and Division of Cardiovascular Medicine, Department of Medicine, Perelman School of Medicine (S.D.), University of Pennsylvania, Philadelphia. Heart Institute, Cincinnati Children's Hospital Medical Center, OH (T.D.R.). Department of Pediatrics, University of Cincinnati College of Medicine, OH (E.M.M.). Department of Cardiology, Rigshospitalet, Copenhagen University Hospital, Denmark (H.B., A.A.R.). Division of Cardiology, Children's Hospital of Philadelphia, PA (J.W.R., K.L.). Cardiology Unit (F.G., I.O.) and Cardiogenetics Unit, Meyer Children's Hospital IRCCS, Florence, Italy. Department of Cardiology, Thorax Center, Cardiovascular Institute, Erasmus MC, Rotterdam, the Netherlands (M.M., P.P.Z.). Department of Cardiology, Boston Children's Hospital, MA (D.A.). Section of Cardiovascular Medicine, Yale School of Medicine, New Haven, CT (R.L., J.C.S.). National Heart & Lung Institute, Imperial College London, United Kingdom (J.W.). Division of Cardiovascular Medicine, Department of Medicine, Stanford University School of Medicine, CA (V.N.P., E. Ashley). Department of Medicine, Center for Undiagnosed Diseases, Stanford University, Stanford University School of Medicine, CA. Garvan Institute of Medical Research and UNSW Sydney, Australia (J.L.). Faculty of Medicine and Health, University of Sydney, New South Wales, Australia (B.G.). Department of Biomedical Engineering, University of Arizona, Tucson (R.L.C., J.T.). Istituto Auxologico Italiano IRCCS, Center for Cardiac Arrhythmias of Genetic Origin and Laboratory of Cardiovascular Genetics, Milano, Italy (L.C.). Department of Medicine and Surgery, University of Milano-Bicocca, Italy (L.C.). Heart Institute (InCor), University of Sao Paulo Medical School, Brazil (A.P.).

Sources of Funding

Funding for the Sarcomeric Human Cardiomyopathy Registry has been provided through unrestricted research grants from BioMarin, Bristol Myers Squibb, Cytokinetics, Lexicon Pharmaceuticals, and Tenaya Therapeutics.

Disclosures

Dr Owens consults for Alexion, Avidity, Bayer, Bristol Myers Squibb, Bridgebio, Braveheart, Cytokinetics, Edgewise, Imbria, Kardigan, Lexeo, Stealth, Tenaya and grant to the institution from BMS. Dr Lakdawala has received unrestricted research support from BMS and Pfizer and modest consulting incomes from Kardigan, Alexion, Cytokinetics, BMS, Pfizer, Gemma, Bayer, and Lexeo. Dr Rosano is a consultant for Astellas, AskBio, Bristol Myers Squibb, Bayer, Merck. Dr Grey is the recipient of a Heart Foundation Future Leader Fellowship and has received honoraria from Bristol Myers Squibb for advisory board and educational seminars. Dr Ingles consultants for Kardigan. Dr Olivetto is supported by the European Union Grant Agreement no. 101137115 (SMASH+HCM [Stratification, Management, and Guidance of Hypertrophic Cardiomyopathy Patients using Hybrid Digital Twin Solutions]); call: HORIZON-HLTH-2023-TOOL05; has received research grants from BMS, Cytokinetics, Amicus, Chiesi, Tenaya, Boston Scientific, and Medtronic; and has received advisory board or speaking fees from BMS, Cytokinetics, Amicus, Chiesi, Bayer, Edgewise, Boston Scientific, Tenaya, Rocket Pharma, Lexeo, Imbria, and Novo Nordisk. Dr Adler is chief scientific officer for Lexeo Therapeutics and has an equity interest in Rocket Pharmaceuticals. Dr Ho reported grants from Alexion, BMS, Cytokinetics, Lexicon, and the US National Institutes of Health and personal fees from Alexion, AstraZeneca, BMS, Cytokinetics, Lexicon, Novo Nordisk, and Sanofi. Dr Day reported grants

from BMS and Cytokinetics during the conduct of the study; grants from Lexicon Pharmaceuticals; and personal fees from Cytokinetics, Lexicon Pharmaceuticals, and Solid Biosciences outside the submitted work. Dr Helms reported grants from Tenaya Therapeutics and personal fees from BMS, Lexeo Therapeutics, and Preload Therapeutics outside the submitted work. Dr Stendahl reported unpaid service as a site investigator for clinical trials for Cytokinetics (aficamten) and BMS (mavacamten) and unpaid participation in research studies with Cytokinetics. Dr Ashley reported being a founder of Candela, Deepcell, Parameter Health, Personalis, and Saturnus; being a founder and nonexecutive director of Svexa; being a nonexecutive director of and holding stock in AstraZeneca; and holding stock in Oxford Nanopore and Pacific Biosciences outside the submitted work. Dr Parikh received personal fees from Lexeo Therapeutics outside the submitted work. The other authors report no conflicts.

Supplemental Material

Tables S1–S15

Figures S1–S16

REFERENCES

- Arbelo E, Protonotarios A, Gimeno JR, Arbustini E, Barriales-Villa R, Basso C, Bezzina CR, Biagini E, Blom NA, De Boer RA, et al; ESC Scientific Document Group. 2023 ESC guidelines for the management of cardiomyopathies. *Eur Heart J*. 2023;44:3503–3626. doi: 10.1093/eurheartj/ehad194
- Ho CY, Day SM, Ashley EA, Michels M, Pereira AC, Jacoby D, Cirino AL, Fox JC, Lakdawala NK, Ware JS, et al. Genotype and lifetime burden of disease in hypertrophic cardiomyopathy: insights from the Sarcomeric Human Cardiomyopathy Registry (SHaRe). *Circulation*. 2018;138:1387–1398. doi: 10.1161/CIRCULATIONAHA.117033200
- Tardiff JC. Thin filament mutations: developing an integrative approach to a complex disorder. Robbins J, Seidman C, Watkins H, eds. *Circ Res*. 2011;108(6):765–782
- Ishida H, Narita J, Ishii R, Sugino H, Tsuru H, Wang R, Yoshihara C, Ueyama A, Ueda K, Hirose M, et al. Clinical outcomes and genetic analyses of restrictive cardiomyopathy in children. *Circ Genom Precis Med*. 2023;16:382–389. doi: 10.1161/CIRCGEN.122.004054
- Coppini R, Ho CY, Ashley E, Day S, Ferrantini C, Girolami F, Tomberli B, Bardi S, Torricelli F, Cecchi F, et al. Clinical phenotype and outcome of hypertrophic cardiomyopathy associated with thin-filament gene mutations. *J Am Coll Cardiol*. 2014;64:2589–2600. doi: 10.1016/j.jacc.2014.09.059
- Pasquale F, Syrris P, Kaski JP, Mogensen J, McKenna WJ, Elliott P. Long-term outcomes in hypertrophic cardiomyopathy caused by mutations in the cardiac troponin T gene. *Circ Cardiovasc Genet*. 2012;5:10–17. doi: 10.1161/CIRCGENETICS.111.959973
- Ommen SR, Mital S, Burke MA, Day SM, Deswal A, Elliott P, Evanovich LL, Hung J, Joglar JA, Kantor P, et al; Writing Committee Members. 2020 AHA/ACC guideline for the diagnosis and treatment of patients with hypertrophic cardiomyopathy: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2020;142:e533–e557. doi: 10.1161/CIR.0000000000000938
- ClinGen. Evidence repository - clinical genome resources
- Larrañaga-Moreira JM, Ochoa JP, Peteiro-Deben R, Martín-Álvarez E, Ripoll-Vera T, Álvarez-Rubio J, Peña-Peña ML, Llamas-Gómez H, Fernández A, Gallego-Delgado M, et al. The p.Asn271Ile variant in the TNNT2 gene is associated with low-risk late-onset hypertrophic cardiomyopathy. *JACC Heart Fail*. 2025;13:102419. doi: 10.1016/j.jchf.2025.01.024
- Sheng JJ, Jin JP. TNNI1, TNNI2 and TNNI3: evolution, regulation, and protein structure–function relationships. *Gene*. 2016;576:385–394. doi: 10.1016/j.gene.2015.10.052
- Zhang Z, Akhter S, Mottl S, Jin J. Calcium-regulated conformational change in the C-terminal end segment of troponin I and its binding to tropomyosin. *FEBS J*. 2011;278:3348–3359. doi: 10.1111/j.1742-4658.2011.08250.x
- Akhter S, Bueltmann K, Huang X, Jin JP. Restrictive cardiomyopathy mutations demonstrate functions of the C-terminal end-segment of troponin I. *Arch Biochem Biophys*. 2014;552:553–560. doi: 10.1016/j.ab.2013.12.001
- Moolman JC, Corfield VA, Posen B, Ngumbela K, Seidman C, Brink PA, Watkins H. Sudden death due to troponin T mutations. *J Am Coll Cardiol*. 1997;29:549–555. doi: 10.1016/s0735-1097(96)00530-x
- Fahed AC, Nemer G, Bitar FF, Arnaout S, Abchee AB, Batrawi M, Khalil A, Abou Hassan OK, DePalma SR, McDonough B, et al. Founder mutation in N terminus of cardiac troponin I causes malignant hypertrophic cardiomyopathy. *Circ Genom Precis Med*. 2020;13:444–452. doi: 10.1161/CIRCGEN.120.002991
- Van Den Wijngaard A, Volders P, Van Tintelen JP, Jongbloed JDH, Van Den Berg MP, Lekanne Deprez RH, Mannens MMAM, Hofmann N, Slegtenhorst M, Dooijes D, et al. Recurrent and founder mutations in the Netherlands: cardiac troponin I (TNNI3) gene mutations as a cause of severe forms of hypertrophic and restrictive cardiomyopathy. *Neth Heart J*. 2011;19:344–351. doi: 10.1007/s12471-011-0135-z
- Cheng Y, Regnier M. Cardiac troponin structure-function and the influence of hypertrophic cardiomyopathy associated mutations on modulation of contractility. *Arch Biochem Biophys*. 2016;601:11–21. doi: 10.1016/j.ab.2016.02.004
- Schober T, Huke S, Venkataraman R, Gryshchenko O, Kryshchal D, Hwang HS, Baudenbacher FJ, Knollmann BC. Myofilament Ca sensitization increases cytosolic Ca binding affinity, alters intracellular Ca homeostasis, and causes pause-dependent Ca-triggered arrhythmia. *Circ Res*. 2012;111:170–179. doi: 10.1161/CIRCGEN.112.270041
- Huke S, Knollmann BC. Increased myofilament Ca²⁺-sensitivity and arrhythmia susceptibility. *J Mol Cell Cardiol*. 2010;48:824–833. doi: 10.1016/j.jmcc.2010.01.011
- Psaras Y, Margara F, Cicconet M, Sparrow AJ, Repetti GG, Schmid M, Steeples V, Wilcox JAL, Bueno-Orovio A, Redwood CS, et al. CalTrack: high-throughput automated calcium transient analysis in cardiomyocytes. *Circ Res*. 2021;129:326–341. doi: 10.1161/CIRCGEN.121.318868
- Wang L, Kim K, Parikh S, Cadar AG, Bersell KR, He H, Pinto JR, Kryshchal DO, Knollmann BC. Hypertrophic cardiomyopathy-linked mutation in troponin T causes myofibrillar disarray and pro-arrhythmic action potential changes in human iPSC cardiomyocytes. *J Mol Cell Cardiol*. 2018;114:320–327. doi: 10.1016/j.jmcc.2017.12.002
- Galińska A, Hatch V, Craig R, Murphy AM, Van Eyk JE, Wang CLA, Lehman W, Foster DB. The C terminus of cardiac troponin I stabilizes the Ca²⁺-activated state of tropomyosin on actin filaments. *Circ Res*. 2010;106:705–711. doi: 10.1161/CIRCGEN.109.210047
- Hoffman RMB, Blumenschein TMA, Sykes BD. An interplay between protein disorder and structure confers the Ca²⁺ regulation of striated muscle. *J Mol Biol*. 2006;361:625–633. doi: 10.1016/j.jmb.2006.06.031
- Zhou Z, Li KL, Rieck D, Ouyang Y, Chandra M, Dong WJ. Structural dynamics of C-domain of cardiac troponin I protein in reconstituted thin filament. *J Biol Chem*. 2012;287:7661–7674. doi: 10.1074/jbc.M111.281600
- O'Mahony C, Jichi F, Ommen SR, Christiaans I, Arbustini E, Garcia-Pavia P, Cecchi F, Olivetto I, Kitaoka H, Gotsman I, et al. International external validation study of the 2014 European Society of Cardiology guidelines on sudden cardiac death prevention in hypertrophic cardiomyopathy (EVIDENCE-HCM). *Circulation*. 2018;137:1015–1023. doi: 10.1161/CIRCULATIONAHA.117.030437
- O'Mahony C, Jichi F, Pavlou M, Monserrat L, Anastasakis A, Rapezzi C, Biagini E, Gimeno JR, Limongelli G, McKenna WJ, et al; Hypertrophic Cardiomyopathy Outcomes Investigators. A novel clinical risk prediction model for sudden cardiac death in hypertrophic cardiomyopathy (HCM Risk-SCD). *Eur Heart J*. 2014;35:2010–2020. doi: 10.1093/eurheartj/ehd439
- Zhao H, Tan Z, Liu M, Yu P, Ma J, Li X, Wang J, Zhao Y, Zhu W, Liu X. Is there a sex difference in the prognosis of hypertrophic cardiomyopathy? A systematic review and meta-analysis. *JAMA*. 2023;12:e026270. doi: 10.1161/jaha.122.026270